

Docker Container — Celery Worker

*N workers
(1 per GPU)*

FITS Image

RabbitMQ
AMQP Broker

GPU (CuPy/CUDA)

- Background Estimation (FFT)
- Cosmetic Filtering (CR/SP)
- Isolated Star Detection
- PSF Modeling (PCA/SVD)
- Source Detection
- Aperture Photometry (FFT)

CPU

- PSF Profile Fitting (Imfit)
- Astrometry (astrometry.net)
- Reference Catalog Query
- Zero-point Calibration
- Catalog Crossmatch (cKDTree / cuML*)

**cuML: GPU-based (adaptive, optional; x86-64 only)*

*GPU Affinity:
\${GPU_ID}*

*OOM Fallback: opt-in
(on_failure / always in JSON config; default: never)*

Outputs
Catalog (FITS) | PostgreSQL+Q3C | Alerts

GPU context reset between tasks

Redis
Result Backend

Logstash
Log Aggregation

